

Estimating Ground Reaction Forces (GRFs) from Inertial Measurement Unit (IMU) sensors

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Viterbi

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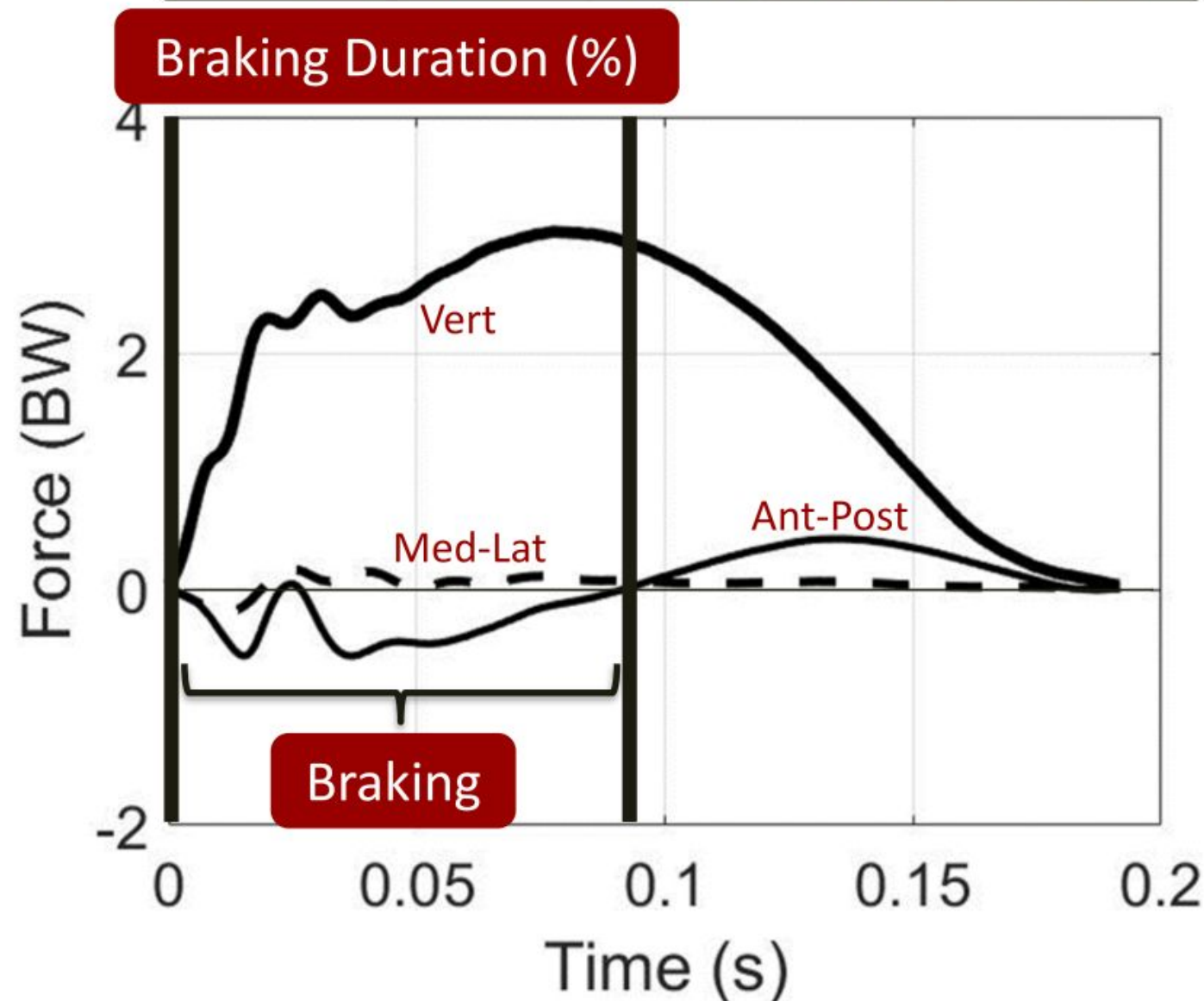
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Objective

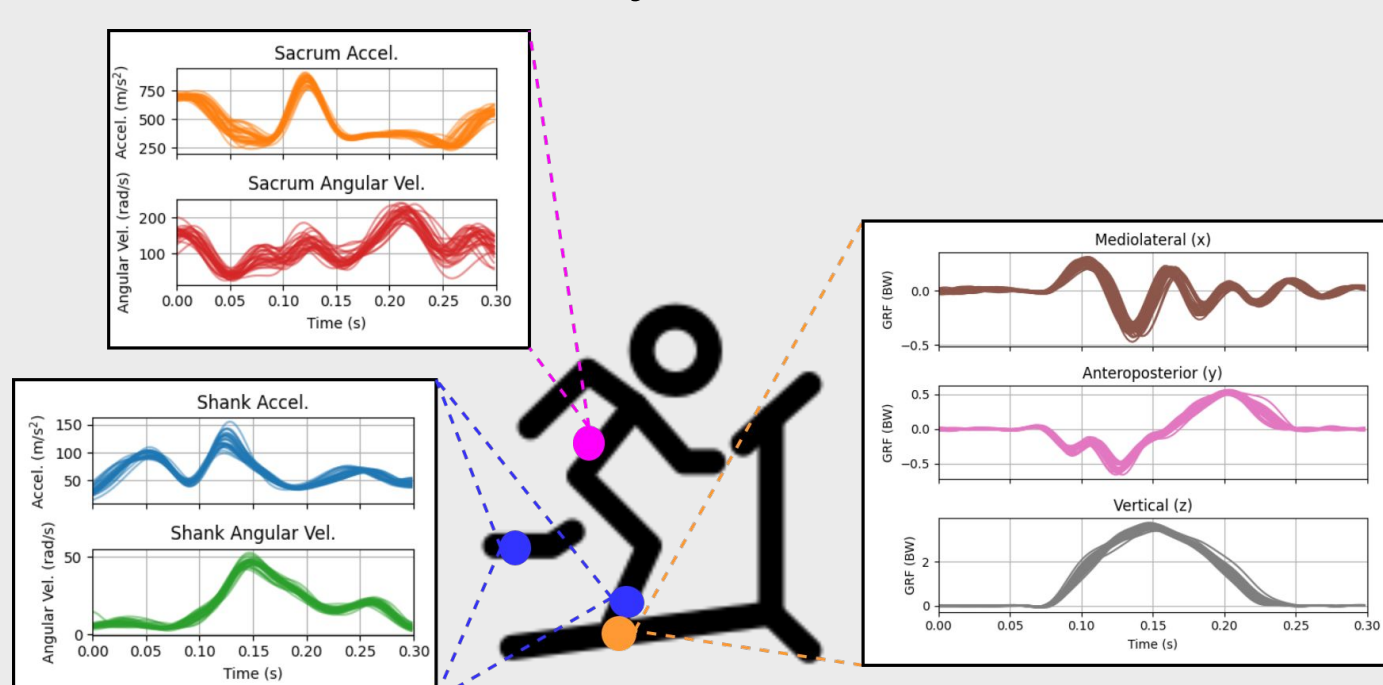
Using data collected with inertial measurement units (IMUs) during running, we aim to estimate ground reaction forces (GRFs) and to derive biomechanical variables (e.g., contact time, impulse, change in velocity) using lightweight machine-learning approaches. In contrast, state-of-the-art estimation using LSTMs suffers from prohibitive inference times on edge devices, requires expensive training and hyperparameter optimization, and results in models that are challenging to interpret.



Athlete performance is often reflected in key biomechanical variables such as: braking duration, loading rate, impulse, net Anterior/Posterior velocity, etc.; these can be driven from GRFs.

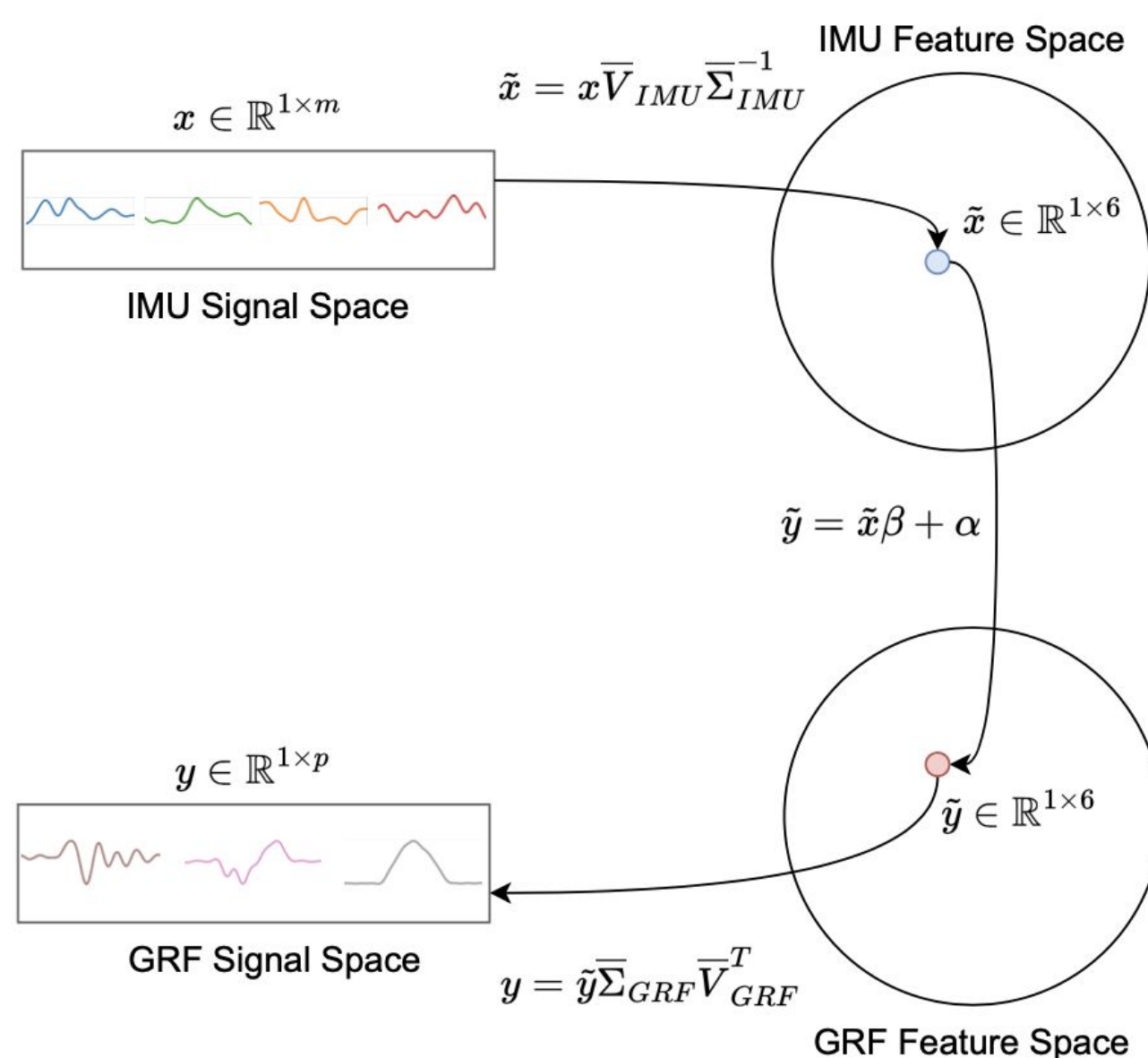
Analysis of GRFs is used to characterize the mechanical loading, which is clinically applicable to identifying athletes at risk for stress-related injuries.

Direct measurement of GRFs is typically performed using force plates or instrumented treadmills in a laboratory environment. Our study evaluates machine learning approaches to estimate GRFs from IMU signals from shanks and sacrum collected on ordinary treadmills.



Method

We propose a novel lightweight solution, SVD Embedding Regression (SER), using linear regression between SVD embeddings of IMU data and GRF data. Through extensive experiments, we illustrate the advantages of lightweight solutions (including SER and k-Nearest-Neighbors (KNN) regression) with state-of-the-art LSTMs.

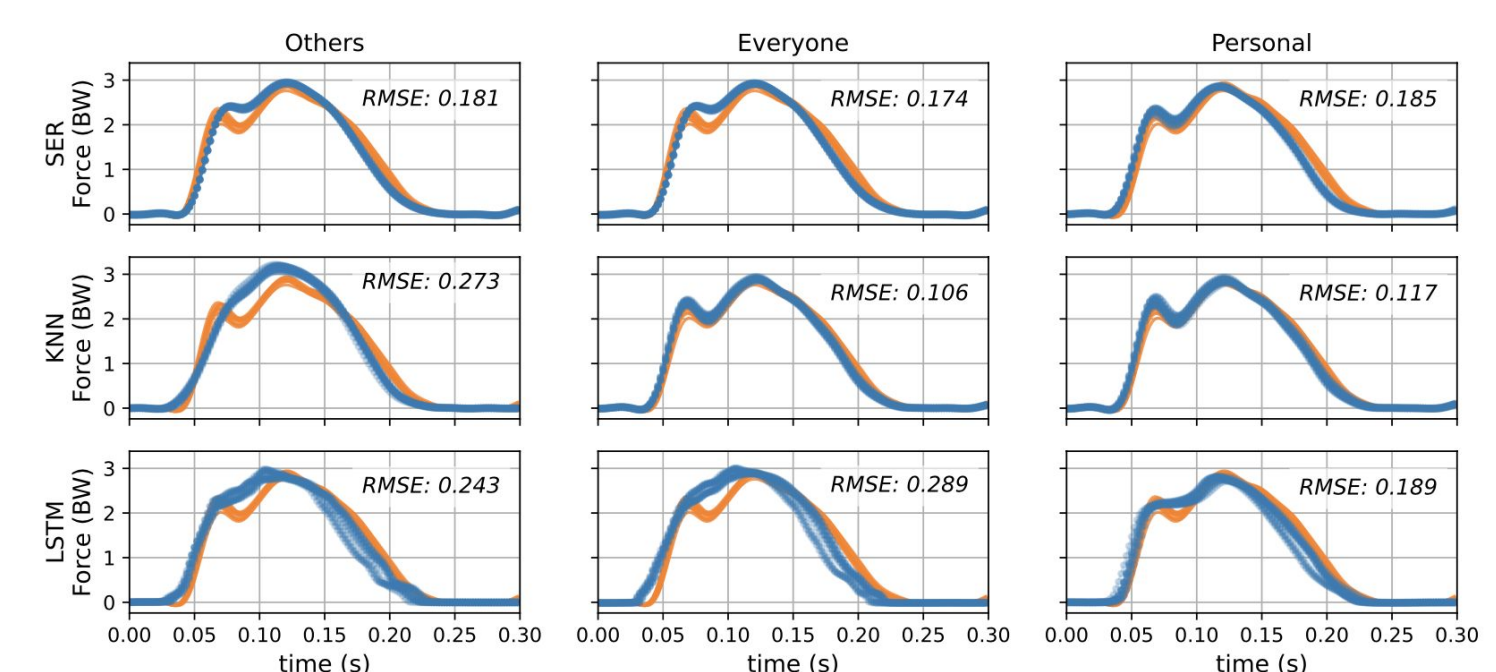


Results

Our experiments are performed under multiple scenarios (including use of training data across multiple athletes vs use of individual athlete data) and combinations of IMU signals (including acceleration and angular velocity) at multiple locations on the body (including sacrum and shanks); we quantify estimation errors for predicting GRFs and biomechanical variables.

Results (cont)

A lack of data can lead LSTMs to produce artifacts. Simple machine learning methods such as SER and KNN are similarly accurate or more accurate than LSTMs, and notably produce more accurate estimations of the GRF waveform when personal training data (i.e., GRF and IMU measurements for particular athlete) are available; in this case, the error of the estimated biomechanical variables is also greatly improved.



Conclusion

Our results illustrate that lightweight solutions such as SER and KNN can be similarly accurate or more accurate than LSTMs. The use of personal data reduces estimation errors of all methods, particularly for most biomechanical variables (as compared to GRFs); moreover, this gain is more pronounced in the lightweight methods.

Song, B., et al. "Estimating Ground Reaction Forces from Inertial Sensors." IEEE Transactions on Biomedical Engineering (2024).

Input Signals	Scenario OTHERS			Scenario EVERYONE			Scenario PERSONAL		
	SER	KNN	LSTM (GPU)	SER	KNN	LSTM (GPU)	SER	KNN	LSTM (GPU)
ALL	0.001	2.256	4.221 (0.497)	0.001	2.088	3.219 (0.472)	0.001	0.042	4.922 (0.558)
ACC	0.003	1.165	4.325 (0.521)	0.002	1.089	3.251 (0.473)	0.001	0.022	5.139 (0.586)
ANG	0.004	1.161	4.605 (0.543)	0.0004	1.064	3.180 (0.474)	0.001	0.025	4.179 (0.516)
SHANK	0.005	1.165	4.014 (0.503)	0.002	1.062	3.216 (0.476)	0.001	0.021	4.739 (0.554)
SACRUM	0.004	1.156	4.557 (0.545)	0.0004	1.067	3.295 (0.480)	0.001	0.023	4.945 (0.563)
SAC/ACC3D	0.001	1.716	3.132 (0.439)	0.002	1.574	5.160 (0.579)	0.001	0.030	0.784 (0.248)
SAC/ACC	0.005	0.518	3.221 (0.450)	0.002	0.411	3.153 (0.461)	0.001	0.013	0.862 (0.272)

TABLE V: Average inference time (in seconds) to estimate 3D GRFs of a collection (120-180 steps) for an athlete. Inference time is measured on an Intel i7-6800K CPU; for LSTM we also report average inference time using a TITAN X GPU.

Input Signals	Scenario OTHERS			Scenario EVERYONE			Scenario PERSONAL		
	SER	KNN	LSTM	SER	KNN	LSTM (FINE-TUNED)	SER	KNN	LSTM
ALL	6.5	6.0	4.2	6.2	3.9	4.2 (3.9)	4.2	4.0	4.3
ACC	7.4	6.6	6.0	7.0	4.1	5.9 (5.2)	4.1	4.1	4.8
ANG	6.5	6.2	6.1	6.3	4.3	6.0 (6.0)	4.3	4.3	5.9
SHANK	7.2	7.0	7.0	6.8	4.9	7.0 (6.7)	4.5	4.4	6.5
SACRUM	7.2	6.9	10.1	6.8	4.2	9.8 (9.6)	4.4	4.5	6.3
SAC/ACC3D	6.4	6.0	5.8	6.2	4.0	5.9 (5.4)	4.1	4.3	6.0
SAC/ACC	6.6	6.3	6.3	6.3	4.3	6.9 (6.1)	4.2	4.4	6.5

TABLE IV: rRMSE (%) of vertical GRF estimations $rRMSE(g_z, \hat{g}_z)$ for different input signals, data scenarios, machine learning methods (results highlighted in blue are optimal or less than 0.5% from optimal for a scenario and set of input signals)